

THE UNIVERSITY OF EDINBURGH

School of Mathematics



Assignment A2:

Course: Operational Research in the Energy Industry

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Submitted To:

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Part A: Operational Problem

1 Sets

G : Set of generators: $G = \{\text{Coal}, \text{CCGT}, \text{Nucl}, \text{Wind}, \text{Solar}\}$

T : Set of time periods (half-hours): $T = \{433, 444, \dots, 1104\}$, treated cyclically (i.e., $|T| = 672$)

2 Parameters

H_t : Duration of each period; $H_t = 0.5$ h for all $t \in T$

PD_t : Power demand in period t (GW)

PW_t : Wind power available in period t (GW), scaled to model capacity

PS_t : Solar power available in period t (GW), scaled to model capacity

PG_g^+ : Installed capacity (Maximum power) of generator g (GW)

P_{CH}^+ : Maximum PHES charging power (GW)

P_{DC}^+ : Maximum PHES discharging power (GW)

Q^+ : PHES energy storage capacity (GWh)

η : PHES round-trip efficiency

CG_g : Operating cost (OPEX) of generator g (£k/GWh of electrical output)

C_{CH} : PHES charging OPEX (£k/GWh of charging input)

U_g : CO₂ intensity of generator g (tCO₂/GWh), (i.e. the standard tCO₂/MWh values $\times 1000$, so that the product $U_g \cdot pG_{gt}$ [tCO₂/GWh $\times$ GW] = [tCO₂/h] directly)

U^+ : Average hourly CO₂ limit (tCO₂/h)

Numerical values:

Parameter	Coal	CCGT	Nucl	Wind	Solar	PHES
PG_g^+ (GW)	25.00	20.00	10.00	20.775	13.146	1.0 (in/out)
CG_g (£k/GWh)	17.9	29.2	15.0	0.0	0.0	2.3 [†]
U_g (tCO ₂ /GWh)	837.0	342.0	0.0	0.0	0.0	—

[†] PHES OPEX $C_{CH} = 2.3$ £k/GWh applies to charging input energy only. $P_{CH}^+ = P_{DC}^+ = 1.0$ GW; $Q^+ = 6.0$ GWh; $\eta = 0.7$.

3 Decision Variables

p_{gt}^G : Power output of generator g in period t (GW)

p_t^{CH} : PHES charging power in period t (GW)

p_t^{DC} : PHES discharging power in period t (GW)

q_t : Energy stored in PHES at end of period t (GWh)

s_t : Generation curtailed in period t (GW)

4 LP Formulation

Objective

The objective is to minimize the average hourly operational cost (OPEX) over the slice as below:

$$\min \quad \frac{H_t}{|T|} \sum_{t \in T} \left[\sum_{g \in G} C_{G_g} \cdot p_{gt}^G + C_{CH} \cdot p_t^{CH} \right] \quad (1)$$

Since $|T|$ is a constant, minimizing the average hourly cost is equivalent to minimizing the total cost over the slice. Wind and Solar have $C_{G_g} = 0$, while shedding carries no cost.

Constraints

Power balance:

$$\sum_{g \in G} p_{gt}^G + p_t^{CH} - p_t^{DC} - s_t = PD_t \quad \forall t \in T \quad (2)$$

The total generation plus the discharge of PHES minus the charging of PHES minus the curtailed generation must equal the demand in every half-hour. The dual variable of this constraint is the marginal cost of supply (£k/GWh), plotted in graph G3 (Task A4).

Generator capacity limits:

$$0 \leq p_{gt}^G \leq PG_g^+ \quad \forall g \in G, t \in T \quad (3)$$

The output of each generator is bounded above by its installed capacity and below by zero (no minimum stable generation in this LP).

Wind availability:

$$0 \leq p_{\text{Wind},t}^G \leq PW_t \quad \forall t \in T \quad (4)$$

The wind output cannot exceed the available resource, scaled from GB 2015 half-hourly data to the model installed capacity of 20.775 GW ($= 1.5 \times$ the 2015 level of 13.85 GW).

Solar availability:

$$0 \leq p_{\text{Solar},t}^G \leq PS_t \quad \forall t \in T \quad (5)$$

The solar output is bounded by the available resource, scaled to the model installed capacity of 13.146 GW. Since, slice-O falls in late January to early March, the solar output is limited.

PHES charging power limit:

$$0 \leq p_t^{CH} \leq P_{CH}^+ \quad \forall t \in T \quad (6)$$

Charging power is bounded by the PHES pump capacity, where $P_{CH}^+ = 1.0$ GW.

PHES discharging power limit:

$$0 \leq p_t^{DC} \leq P_{DC}^+ \quad \forall t \in T \quad (7)$$

Discharging power is bounded by the PHES turbine capacity, where $P_{DC}^+ = 1.0$ GW.

PHES energy balance (cyclic):

$$q_t = q_{\text{nxt}(t)} + \eta \cdot H_t \cdot p_t^{CH} - H_t \cdot p_t^{DC} \quad \forall t \in T \quad (8)$$

where; $\text{nxt}(t) = T_B + (t - T_B + 1) \bmod (T_E - T_B + 1)$, $T_B = 433$, $T_E = 1104$.

Stored energy at the end of period t equals stored energy at the end of the next period, plus energy added by charging (at efficiency $\eta = 0.7$), minus energy removed by discharging. The cyclic boundary condition makes the solution self-consistent over the slice without needing a fixed initial stored energy.

PHES energy capacity:

$$0 \leq q_t \leq Q^+ \quad \forall t \in T \quad (9)$$

Stored energy must stay within the reservoir capacity $Q^+ = 6.0$ GWh at all times.

CO₂ emissions limit (For Task A2–A3):

$$\frac{H_t}{|T|} \sum_{t \in T} (U_{\text{Coal}} \cdot p_{\text{Coal},t}^G + U_{\text{CCGT}} \cdot p_{\text{CCGT},t}^G) \leq U^+ \quad (10)$$

(Nuclear, Wind and Solar have $U_g = 0$ and do not appear.)

Average hourly CO₂ must not exceed U^+ (tCO₂/h). The dual variable κ of this constraint is the marginal cost of relaxing the limit by 1 tCO₂/h, interpretable as an implicit carbon price (£k/tCO₂).

Non-negativity and domain constraints:

$$p_{gt}^G, p_t^{CH}, p_t^{DC}, q_t, s_t \geq 0 \quad \forall g \in G, t \in T \quad (11)$$

All decision variables are physically non-negative. In Mosel these are satisfied implicitly since `mpvar` variables have a default lower bound of zero.

5 Results and Discussion

Task A1: Finding $U_{\max}$

To find $U_{\max}$ we solve (1) subject to constraints all constraints apart from CO₂ limit. After obtaining the optimal dispatch, we compute:

$$U_{\max} = \frac{H_t}{|T|} \sum_{t \in T} (U_{\text{Coal}} \cdot p_{\text{Coal},t}^{G*} + U_{\text{CCGT}} \cdot p_{\text{CCGT},t}^{G*})$$

This is the emissions level of the least-cost dispatch and is the upper end of the parametric range in Task A3. After running the model the following is the summary of results obtained with no CO₂ limit constraint.

Results for $U_{\max}$

Quantity	Value	Units
$U_{\max}$ (avg hourly CO ₂)	7,272.20	tCO ₂ /h
Avg hourly OPEX	247.754	£k/h
Avg Coal output	16.712	GW
Avg CCGT output	1.627	GW
Avg Nuclear output	9.894	GW
Avg Wind output / available	9.842 / 9.842	GW
Avg Solar output / available	0.325 / 0.325	GW
Avg PHES charge	0.191	GW
Avg PHES discharge	0.134	GW
Total generation shed	0	GWh
Total energy stored (cyclic)	44.95	GWh
Storage min / max	0 / 6.0	GWh
Storage swing	6.0	GWh

From the results we can see that the unconstrained merit order is Nuclear (£15 k/GWh) < Coal (£17.9 k/GWh) < CCGT (£29.2 k/GWh), so Nuclear runs near full capacity (9.89 GW) and Coal dominates the residual load. CCGT provides only 1.63 GW on average. All available wind and solar is absorbed with zero curtailment, and PHES cycles through its full 6 GWh swing to arbitrage cheap off-peak generation to higher-cost peak periods.

Task A2: Finding $U_{\min}$

We replace objective (1) with the average hourly CO₂ expression and minimize subject to all other constraints. The OPEX of this solution is recorded for comparison but is not minimized. The table below shows the results for $U_{\min}$, and the recorded OPEX.

Results $U_{\min}$

Quantity	Value	Units
$U_{\min}$ (avg hourly CO ₂)	3,954.82	tCO ₂ /h
Avg hourly OPEX (not minimised)	323.641	£k/h

To minimize CO₂, the model displaces Coal (837 tCO₂/GWh) with CCGT (342 tCO₂/GWh), raising average OPEX by £75.9 k/h (+30.6 %) relative to Task A1. The residual 3,954.8 tCO₂/h reflects CO₂ from the Coal and CCGT still required to meet demand after all Nuclear, Wind and Solar are dispatched. Further reductions would require additional low-carbon capacity.

Task A3: Parametric CO₂ sweep

For each $\alpha \in \{1.0, 0.75, 0.5, 0.25, 0.0\}$, we set

$$U^+(\alpha) = \alpha U_{\max} + (1 - \alpha) U_{\min}$$

activate constraint (10) at $U^+(\alpha)$, and minimise the OPEX objective (1). The shadow price κ of (10) gives the implicit carbon price at each point on the cost emissions trade-off curve. The following results were obtained in this case:

Results Obtained

Table 1: Task A3: parametric CO₂ sweep. $U_{\max} = 7,272.2 \text{ tCO}_2/\text{h}$; $U_{\min} = 3,954.8 \text{ tCO}_2/\text{h}$.

Quantity	$\alpha = 1.0$	$\alpha = 0.75$	$\alpha = 0.5$	$\alpha = 0.25$	$\alpha = 0.0$
(1) U^+ (tCO ₂ /h)	7272.2	6442.9	5613.5	4784.2	3954.8
(2) κ (£k/tCO ₂)	0	-0.0228	-0.0228	-0.0228	-0.0778
(3) Avg OPEX (£k/h)	247.75	265.47	284.41	303.34	323.64
(4) Avg Coal (GW)	16.712	13.409	10.058	6.707	3.314
(4) Avg CCGT (GW)	1.627	4.862	8.213	11.563	15.018
(4) Avg Nucl (GW)	9.894	9.923	9.923	9.923	9.923
(4) Avg Wind (GW)	9.842	9.842	9.842	9.842	9.842
(4) Avg Solar (GW)	0.325	0.325	0.325	0.325	0.325
(4) Avg PHES chg (GW)	0.191	0.057	0.057	0.057	0.261
(4) Avg PHES dch (GW)	0.134	0.040	0.040	0.040	0.183
(5) Storage swing (GWh)	6.00	6.00	6.00	5.43	6.00
(6) Avg shed (GW)	0.00	0.00	0.00	0.00	0.00

Key observations:

- **Coal to CCGT substitution:** As α falls from 1 to 0, average Coal output drops from 16.7 GW to 3.3 GW while CCGT rises from 1.6 GW to 15.0 GW. Total thermal output (Coal + CCGT) stays approximately constant because demand and renewable availability are unchanged.
- **Shadow price κ :** For $\alpha \in \{0.75, 0.5, 0.25\}$, κ is constant at -0.0228 £k/tCO_2 . This value can be derived analytically as the cost per unit of CO₂ abated by switching from Coal to CCGT:

$$|\kappa| = \frac{CG_{\text{CCGT}} - CG_{\text{Coal}}}{U_{\text{Coal}} - U_{\text{CCGT}}} = \frac{29.2 - 17.9}{837 - 342} \approx 0.0228 \text{ £k/tCO}_2$$

confirming the LP dual value. At $\alpha = 0$ the constraint is very tight and κ jumps to -0.0778 £k/tCO_2 , indicating sharply increasing marginal abatement cost once the simple Coal to CCGT swap is exhausted.

- **OPEX cost:** Hourly OPEX rises from £247.75 k/h to £323.64 k/h (+30.6 %), driven entirely by the fuel switch from Coal to CCGT.
- **No generation shedding:** Curtailment is zero at all α values showing that the dispatchable plant and PHES can always balance supply and demand in Slice-O.
- **Storage utilisation:** PHES swings through its full 6 GWh at all α values except $\alpha = 0.25$ (5.43 GWh). Charging and discharging activity is highest at $\alpha = 0$, where the tighter CO₂ constraint creates larger inter-period price differentials and greater incentive for storage arbitrage.

- **Nuclear, Wind and Solar are unaffected:** All three zero-CO₂ generators run at essentially constant average output across the sweep, confirming they are always dispatched ahead of any fossil plant (do not depend on CO₂ constraint).

Task A4: Dispatch graphs for $\alpha = 0.5$

In this part we re solve the model at $\alpha = 0.5$ ($U^+ = 5,613.5 \text{ tCO}_2/\text{h}$) and extract the period-by-period solution for periods 433 to 1104. Three graphs are produced from the exported dispatch data are as follows:

- **G1 - Stacked generation area plot:** The figure (1) below shows the demand (PD_t) as the upper reference and the generation stacked from bottom in merit order (Nuclear, Wind, Solar, Coal, CCGT). The generation shed (s_t) is not shown since it is zero in this case. Nuclear runs as a flat baseload at $\sim 10 \text{ GW}$ throughout, never varying, as expected for the cheapest zero-CO₂ generator. Wind dominates the variable layer, driving large periodic swings with peaks reaching $\sim 30 \text{ GW}$ above the nuclear base and troughs near zero, reflecting GB winter intermittency.

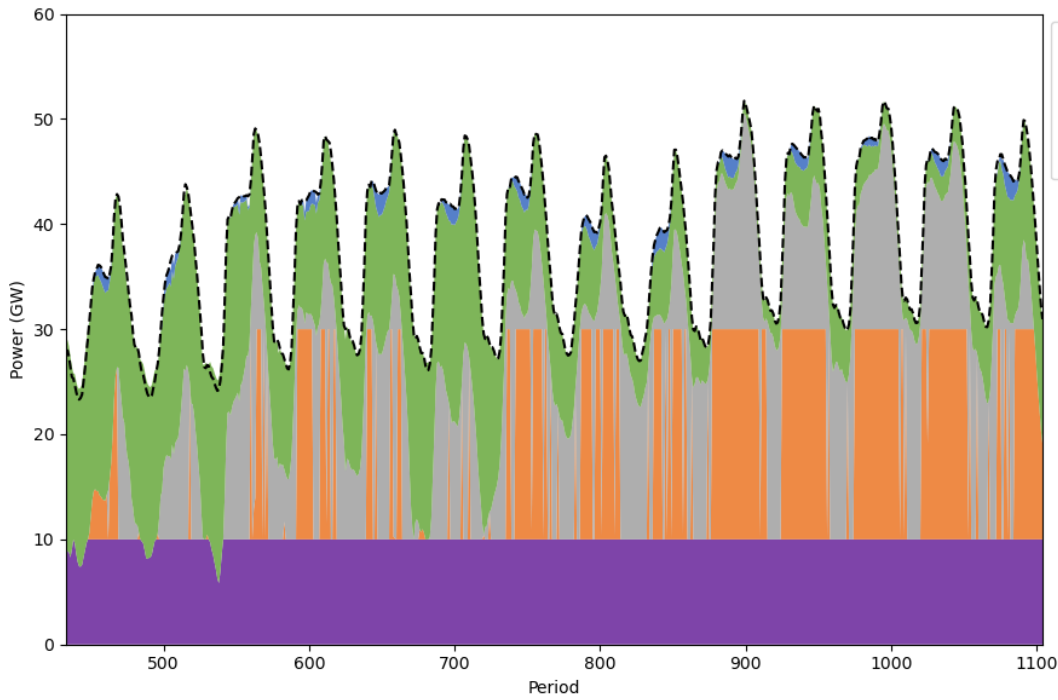


Figure 1: Stacked Generation Power Plot For $\alpha = 0.5$

- **G2 - Storage energy:** The figure (2) shows the energy level (q_t) versus the period t (GWh). Additionally I have plotted the power difference (Discharging - Charging) in the PHES. PHES activity is concentrated in periods 443 to 700, where the storage level cycles between 0 and 6 GWh and power hits both 1 GW bounds frequently. After period 700 the stored energy stabilizes near 4 GWh with negligible cycling, indicating that there is no price differential worth exploiting. This suggests that the latter part of Slice O has a well balanced dispatch where thermal generation alone can follow demand without storage.

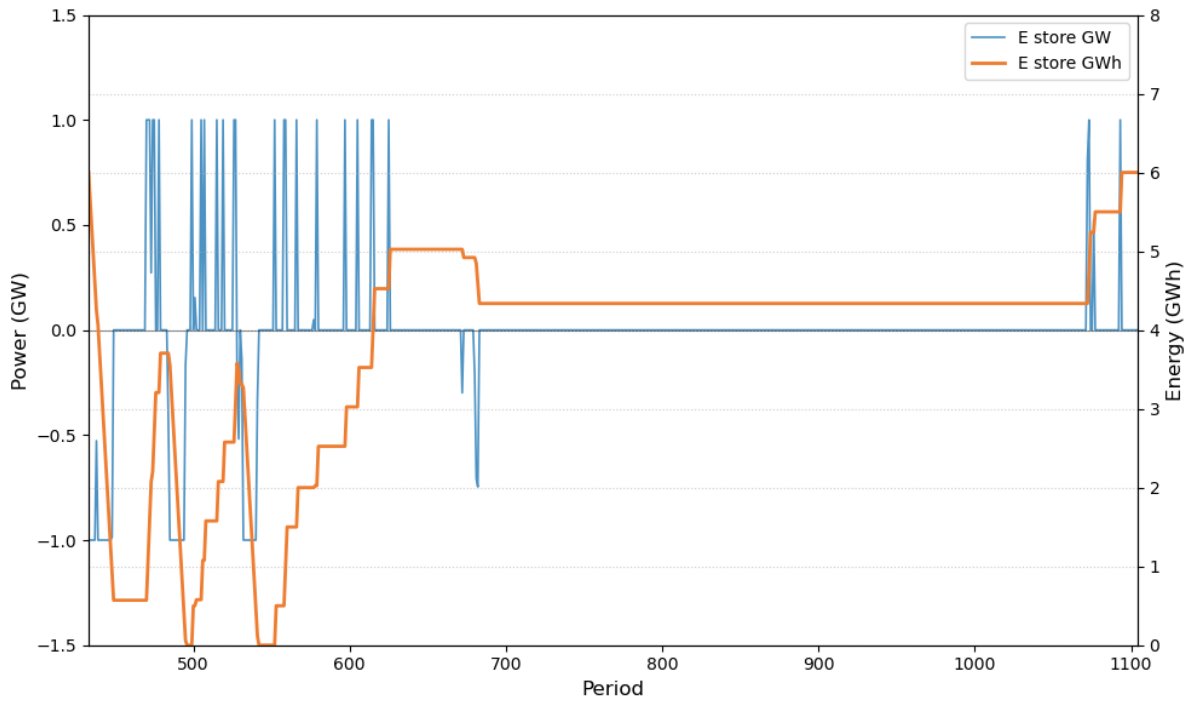


Figure 2: The energy in the electricity store

- G3 - Marginal cost:** The figure (3) below shows the dual of constraint of power balance against the period t (£k/GWh). The marginal cost takes only three discrete levels (£11.4, £17.9 and £27.8/MWh). Brief dips to lower values in periods 443 to 700 correspond exactly to the active PHES periods in G2, where discharging displaces CCGT and reduces the system marginal price.

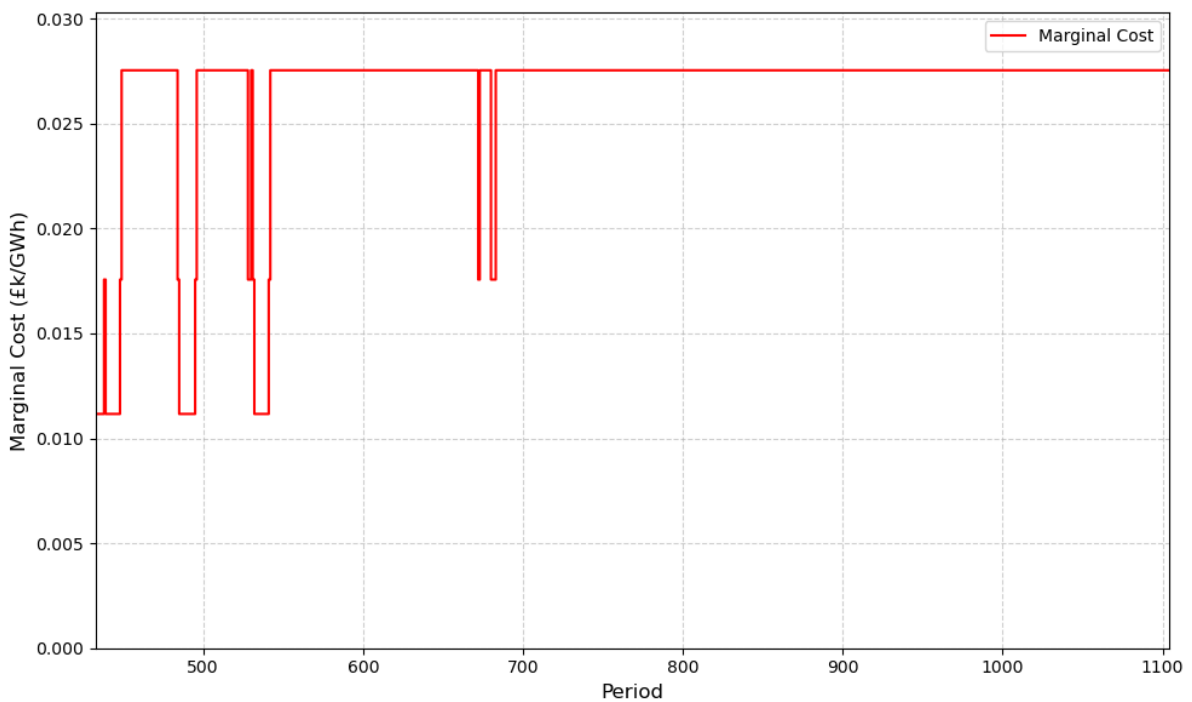


Figure 3: The marginal cost of extra demand

Task A5: Comment on results

The key results from Tasks A1 to A4 have been discussed in detail in their respective sections above. The following briefly highlights the most interesting and surprising features.

The most notable result is that the **shadow price κ is constant at -0.0228 £k/tCO₂ across three of the five α values** ($\alpha = 0.75, 0.5, 0.25$), confirming analytically that the marginal abatement cost is entirely determined by the fixed cost differential between Coal and CCGT. The jump to -0.0778 £k/tCO₂ at $\alpha = 0$ is striking in magnitude, revealing that the final increment of abatement is $3.4\times$ more expensive than all preceding increments.

A second interesting feature is that **no generation shedding occurs at any α value** despite fixed renewable capacities and a small 1 GW PHES, indicating that wind and solar penetration in the Part A system is never high enough to create surplus generation.

Finally, the **PHES activity in G2 is more localized than expected**: rather than cycling continuously throughout the 672-period slice, storage is active almost exclusively in periods 433 to 700, suggesting that the marginal cost differentials that motivate arbitrage are concentrated in a specific meteorological window rather than spread evenly across the slice.

Part B: Joint Investment–Operational Problem

6 Sets

G : Set of generators: $G = \{\text{Coal}, \text{CCGT}, \text{Nucl}, \text{Wind}, \text{Solar}\}$

T : Set of time periods (half-hours): $T = \{1, 2, \dots, 17520\}$, treated cyclically ($|T| = 17520$, full GB 2015 year)

7 Parameters

All Part A parameters carry over. The following are added or modified for Part B:

W_t : Per-unit wind availability in period t , defined as $W_t = \frac{\text{WinMW}(t)/1000}{\bar{P}_W^{2015}}$ where $\bar{P}_W^{2015} = 13.85$ GW is the 2015 installed wind capacity. Available wind power $= P_W \cdot W_t$ (GW).

SR_t : Per-unit solar availability in period t , defined as $SR_t = \frac{\text{SolMW}(t)/1000}{\bar{P}_S^{2015}}$ where $\bar{P}_S^{2015} = 8.76$ GW. Available solar power $= P_S \cdot S_t$ (GW).

$\bar{P}_g$: Fixed capacity of $g \in \{\text{Coal}, \text{CCGT}, \text{Nucl}\}$ (GW), same values as PG_g^+ in Part A

P_W^- : Lower bound on wind capacity; $P_W^- = 20.775$ GW

P_S^- : Lower bound on solar capacity; $P_S^- = 13.146$ GW

P_P^- : Lower bound on PHES power capacity; $P_P^- = 1.0$ GW

Q^- : Lower bound on PHES energy capacity; $Q^- = 6.0$ GWh.

K_g^{CPX} : Hourly CAPEX of generator g (£k/GW/h).

K_P^{CPX} : Hourly CAPEX of PHES power; $K_P^{CPX} = 9.3$ £k/GW/h.

K_Q^{CPX} : Hourly CAPEX of PHES energy store; $K_Q^{CPX} = 0.087$ £k/GWh/h.

Numerical values:

Parameter	Coal	CCGT	Nucl	Wind	Solar	PHES
$\bar{P}_g$ / bounds (GW)	fixed 25.0	fixed 20.0	fixed 10.0	≥ 20.775	≥ 13.146	≥ 1.0 (power)
CG_g (£k/GWh)	17.9	29.2	15.0	0.0	0.0	2.3 [†]
U_g (tCO ₂ /GWh)	837.0	342.0	0.0	0.0	0.0	—
K_g^{CPX} (£k/GW/h)	9.9	4.0	20.4	11.7	3.9	9.3 (power)

[†] $CG_{CH} = 2.3$ £k/GWh on charging input only. $K_Q^{CPX} = 0.087$ £k/GWh/h; PHES energy lower bound $Q^- = 6.0$ GWh.

8 Decision Variables

All Part A operational variables carry over: $p_{gt}^G, p_t^{CH}, p_t^{DC}, q_t, s_t$. The following four investment variables are added and solved jointly with the operational variables in a single LP:

P_W : Installed wind power capacity (GW)

P_S : Installed solar power capacity (GW)

P_P : PHES power capacity, charge and discharge equal (GW)

Q : PHES energy storage capacity (GWh)

9 LP Formulation

Objective

Minimize the average hourly total cost (CAPEX + OPEX):

$$\min \quad K_W^{CPX} P_W + K_S^{CPX} P_S + K_P^{CPX} P_P + K_Q^{CPX} Q + \frac{H_t}{|T|} \sum_{t \in T} \left[\sum_{g \in G} C_{Gg} \cdot p_{gt}^G + C_{CH} \cdot p_t^{CH} \right] \quad (12)$$

The first four terms are the annualised hourly CAPEX of the investable equipment. Coal, CCGT and Nuclear CAPEX is constant (fixed capacities) and does not affect the optimal decision, so it is excluded from the objective.

Constraints

Power balance:

$$\sum_{g \in G} p_{gt}^G + p_t^{DC} - p_t^{CH} - s_t = PD_t \quad \forall t \in T \quad (13)$$

The Total generation plus PHES discharge minus PHES charging minus curtailed generation must equal demand in every half-hour. The dual variable is the marginal cost of supply (£k/GWh), used in graph G3 (Task B5).

Thermal generator limits (for fixed capacities):

$$0 \leq p_{gt}^G \leq \bar{P}_g \quad \forall g \in \{\text{Coal, CCGT, Nucl}\}, t \in T \quad (14)$$

Coal, CCGT and Nuclear outputs are bounded by their fixed installed capacities, which cannot be changed by the investment decision.

Wind availability:

$$0 \leq p_{\text{Wind},t}^G \leq P_W \cdot W_t \quad \forall t \in T \quad (15)$$

Wind output is bounded by the installed capacity P_W multiplied by the per-unit availability factor $W_t \in [0, 1]$ derived from the 2015 GB half-hourly wind data. Because W_t is a fixed parameter and P_W is the decision variable, this constraint is linear. The solver chooses P_W optimally, the wind availability shape is determined by the data.

Solar availability:

$$0 \leq p_{\text{Solar},t}^G \leq P_S \cdot SR_t \quad \forall t \in T \quad (16)$$

Analogous to wind availability constraint, for solar. SR_t is the per-unit solar availability from the data.

PHES charging power limit:

$$0 \leq p_t^{CH} \leq P_P \quad \forall t \in T \quad (17)$$

Charging power is bounded by the optimally chosen PHES power capacity P_P (rather than the fixed P_{CH}^+ of Part A).

PHES discharging power limit:

$$0 \leq p_t^{DC} \leq P_P \quad \forall t \in T \quad (18)$$

Discharging power is also bounded by P_P ; charge and discharge capacity are assumed equal.

PHES energy balance (cyclic):

$$q_t = q_{\text{nxt}(t)} + \eta \cdot H_t \cdot p_t^{CH} - H_t \cdot p_t^{DC} \quad \forall t \in T \quad (19)$$

where $\text{nxt}(t) = 1 + (t \bmod 17520)$ (i.e. period 17520 wraps to period 1).

Same cyclic energy balance as Part A but now over the full year. The cyclic condition enforces that the storage level at the start and end of the year are equal, making the solution suitable for long run planning.

PHES energy capacity:

$$0 \leq q_t \leq Q \quad \forall t \in T \quad (20)$$

Stored energy is bounded above by the optimally chosen energy capacity Q (rather than the fixed Q^+ of Part A).

Capacity variables lower bounds:

$$P_W \geq P_W^-, \quad P_S \geq P_S^-, \quad P_P \geq P_P^-, \quad Q \geq Q^- \quad (21)$$

Investment variables must be at least as large as the Part A operational values. This ensures Part B solutions are always at least as capable as the Part A system and allows direct comparison between the two parts.

CO₂ emissions limit (For Tasks B2–B3 only):

$$\frac{H_t}{|T|} \sum_{t \in T} (U_{\text{Coal}} \cdot p_{\text{Coal},t}^G + U_{\text{CCGT}} \cdot p_{\text{CCGT},t}^G) \leq U^+ \quad (22)$$

Average hourly CO₂ must not exceed U^+ (tCO₂/h). The dual variable κ is the implicit carbon price (£k/tCO₂). In Task B1 this constraint is inactive; in Task B2 CO₂ is minimized directly; in Task B3 U^+ is parametrically tightened.

Non-negativity and domain constraints:

$$p_{gt}^G, p_t^{CH}, p_t^{DC}, q_t, s_t, P_W, P_S, P_P, Q \geq 0 \quad \forall g \in G, t \in T \quad (23)$$

All decision variables are physically non-negative. The investment variables are continuous; the lower bounds in (21) together with these non-negativity constraints define the feasible domain.

10 Results and Discussion

Task B1: Finding $U_{\max}$

Same as in part A, we solve (12) subject to all constraints except the CO₂ limit constraint. After obtaining the optimal solution, we compute:

$$U_{\max} = \frac{H_t}{|T|} \sum_{t \in T} (U_{\text{Coal}} \cdot pG_{\text{Coal},t}^* + U_{\text{CCGT}} \cdot pG_{\text{CCGT},t}^*)$$

$U_{\max}$ is the emissions level of the least-cost investment and dispatch plan and forms the upper end of the parametric range in Task B3. The following results were obtained from Mosel code.

Results for $U_{\max}$

Quantity	Value	Units
$U_{\max}$ (avg hourly CO ₂)	5,926.75	tCO ₂ /h
Avg hourly total cost	509.100	£k/h
Avg hourly CAPEX	304.159	£k/h
Avg hourly OPEX	204.941	£k/h
Wind capacity P_W^*	20.775	GW (at lower bound)
Solar capacity P_S^*	13.146	GW (at lower bound)
PHES power capacity P_P^*	1.000	GW (at lower bound)
PHES energy capacity Q^*	6.000	GWh (at lower bound)
Avg Coal output	14.037	GW
Avg CCGT output	0.306	GW
Avg Nuclear output	9.966	GW
Avg Wind output / available	6.414 / 6.414	GW
Avg Solar output / available	1.340 / 1.340	GW
Avg PHES charge	0.083	GW
Avg PHES discharge	0.058	GW
Total generation shed	0	GWh
Total energy stored (cyclic)	508.69	GWh
Storage min / max	0 / 6.0	GWh
Storage swing	6.0	GWh

At $U_{\max}$ the investment variables all sit at their lower bounds: the cheapest solution simply retains the Part A installed capacities with no additional investment. The full-year average output differs from Part A's Slice-O values because Slice-I covers all seasons; in particular, wind and solar averages are lower (6.41 GW and 1.34 GW respectively) due to the inclusion of summer periods with less wind but more solar. Nuclear runs close to its 10 GW ceiling, Coal

dominates the residual thermal load, and CCGT is barely used. Zero shedding confirms that at the lower-bound capacities, all renewable output can be absorbed.

Task B2: Finding $U_{\min}$

We replace the OPEX objective (12) with the average hourly CO_2 expression and minimize subject to all constraints except the CO_2 one. The solver is free to invest in as much wind, solar and PHES as needed (resulting to zero $U_{\min}$). The total cost of this solution is recorded but not minimized.

Results for $U_{\min}$

Quantity	Value	Units
$U_{\min}$ (avg hourly CO_2)	0	t CO_2 /h
Avg hourly total cost (not minimized)	9,014.01	£k/h
Avg hourly CAPEX	8,940.15	£k/h
Avg hourly OPEX	73.851	£k/h
Wind capacity P_W^*	758.85	GW
Solar capacity P_S^*	13.146	GW (at lower bound)
PHES power capacity P_P^*	1.058	GW
PHES energy capacity Q^*	6.000	GWh (at lower bound)
Avg Coal output	0.000	GW
Avg CCGT output	0.000	GW
Avg Nuclear output	9.844	GW
Avg Wind output / available	234.17 / 234.30	GW
Avg Solar output / available	1.340 / 1.340	GW
Avg PHES charge	0.017	GW
Avg PHES discharge	0.012	GW
Total generation shed	1,868,641	GWh
Storage swing	4.44	GWh

$U_{\min} = 0$ is achieved by building enough wind capacity (758.85 GW) to completely displace all fossil generation. Nuclear still runs at 9.84 GW (it has zero CO_2 and low OPEX so there is no incentive to curtail it). The enormous wind overbuild creates massive surplus generation. We get 1,868,641 GWh of curtailment over the year (versus annual GB demand of roughly 160 000 GWh), which is physically unrealistic but an artefact of the LP having no constraint on maximum investment. Solar and PHES remain at or near their lower bounds because additional solar or storage investment does not further reduce CO_2 once coal and CCGT are already zero. The total cost of £9,014 k/h is dominated almost entirely by wind CAPEX.

Task B3: Parametric CO₂ sweep

For each $\alpha \in \{1.0, 0.4, 0.2, 0.1, 0.05, 0.0\}$, we set

$$U^+(\alpha) = \alpha U_{\max} + (1 - \alpha) U_{\min} = \alpha \times 5,926.75$$

(since $U_{\min} = 0$), activate constraint (C10) at $U^+(\alpha)$, and minimize the total cost objective (12). The shadow price κ of CO₂ constraint gives the implicit carbon price at each point on the cost emissions graph.

Results for Task B3

Tables 2 and 3 give the operational and investment results respectively.

Table 2: Task B3: operational results. $U_{\max} = 5,926.75$ tCO₂/h; $U_{\min} = 0$ tCO₂/h.

Quantity	$\alpha = 1.0$	$\alpha = 0.4$	$\alpha = 0.2$	$\alpha = 0.1$	$\alpha = 0.05$	$\alpha = 0.0$
(1) U^+ (tCO ₂ /h)	5926.7	2370.7	1185.3	592.7	296.3	0
(2) κ (£k/tCO ₂)	0	-0.136	-0.309	-0.519	-0.883	-926.1
(3) Avg OPEX (£k/h)	204.94	256.66	158.81	109.16	75.34	27.20
(4) Avg Coal (GW)	14.037	0.747	0.207	0.021	0.002	0.000
(4) Avg CCGT (GW)	0.306	12.036	6.425	3.415	1.727	0.000
(4) Avg Nucl (GW)	9.966	9.873	8.304	7.476	6.101	2.618
(4) Avg Wind (GW)	6.414	8.091	14.520	18.057	21.441	29.215
(4) Avg Solar (GW)	1.340	1.344	3.254	4.692	5.402	5.732
(4) Avg PHES chg (GW)	0.083	0.176	0.762	2.644	3.773	6.574
(4) Avg PHES dch (GW)	0.058	0.123	0.533	1.851	2.641	4.602
(5) Storage swing (GWh)	6.00	6.50	82.72	490.68	1023.4	2484.1
(6) Avg shed (GW)	0.000	0.000	0.443	0.829	1.503	3.554

Table 3: Task B3: investment results.

Quantity	$\alpha = 1.0$	$\alpha = 0.4$	$\alpha = 0.2$	$\alpha = 0.1$	$\alpha = 0.05$	$\alpha = 0.0$
(1) Total cost (£k/h)	509.10	624.57	864.59	1097.24	1302.13	1910.02
(2) CAPEX (£k/h)	304.16	367.91	705.77	988.09	1226.80	1882.82
(3) Wind capacity (GW)	20.775	26.207	47.029	58.484	69.443	94.622
(4) Solar capacity (GW)	13.146	13.186	31.927	46.033	53.005	56.237
(5) PHES power (GW)	1.000	1.000	2.561	8.775	12.748	36.593
(6) PHES energy (GWh)	6.000	6.497	82.721	490.680	1023.44	2484.08

Key observations:

- **Investment grows sharply with tighter CO₂ limits.** At $\alpha = 1.0$ all investment variables sit at their lower bounds (the Part A system suffices). As α decreases, wind, solar and PHES capacity all increase substantially. At $\alpha = 0$ wind reaches 94.6 GW (4.6×

the lower bound), solar 56.2 GW (4.3 $\times$), PHES power 36.6 GW (36.6 $\times$) and PHES energy 2,484 GWh (414 $\times$).

- **OPEX is non-monotonic.** Average OPEX falls from £204.94 k/h at $\alpha = 1$ to £27.20 k/h at $\alpha = 0$, as fossil generation is replaced by zero-marginal-cost wind and solar. However, OPEX rises at $\alpha = 0.4$ (£256.66 k/h) before declining again, because the relatively tight constraint at $\alpha = 0.4$ forces a large dispatch of expensive CCGT to meet residual demand while wind investment has not yet grown enough to compensate.
- **Total cost rises monotonically.** Total cost increases from £509.10 k/h at $\alpha = 1$ to £1,910.02 k/h at $\alpha = 0$ (+275 %), driven by rapidly growing CAPEX. CAPEX accounts for 60% of total cost at $\alpha = 1$ and 99% at $\alpha = 0$, reflecting the shift from an OPEX-dominated to a CAPEX-dominated system.
- **Shadow price κ and the carbon price curve.** κ starts at 0 (non-binding), increases in magnitude to -0.519 £k/tCO₂ at $\alpha = 0.1$, and then jumps dramatically to -926.1 £k/tCO₂ at $\alpha = 0$. This convex shape of the abatement cost curve reflects the fact that early reductions are relatively cheap (substituting coal with wind and CCGT), while achieving zero CO₂ requires a very large overbuild of intermittent renewables that must shed most of their output.
- **Generation shedding appears from $\alpha = 0.2$ onwards.** At $\alpha = 0.2$, average curtailment is 0.443 GW (total 3,884 GWh/year), rising to 3.554 GW (31,130 GWh/year) at $\alpha = 0$. Shedding is necessary because the large renewable overbuild required to meet tight CO₂ limits produces surplus energy that the fixed demand and limited PHES capacity cannot absorb.
- **Nuclear reduces at low α .** Average nuclear output falls from 9.97 GW ($\alpha = 1$) to 2.62 GW ($\alpha = 0$). Although nuclear has zero CO₂, the extremely large wind and solar investment at $\alpha = 0$ creates so much surplus generation that even nuclear must be curtailed at times to maintain the power balance.
- **PHES grows rapidly to manage variability.** The PHES energy capacity grows from 6 GWh to 2,484 GWh across the α values, and PHES power from 1 GW to 36.6 GW. This reflects the increasing need to store excess renewable energy for later use as the renewable share rises.

Task B4: Storage energy plot, $\alpha = 0.0, 0.05, 0.1$

We re-solve the model at $\alpha \in \{0.0, 0.05, 0.1\}$ and record q_t for all $t \in T$ (the full year). Plot the three q_t profiles on a single graph (G2) to illustrate how storage utilisation changes as the CO₂ limit tightens.

Results for B4

The storage time series were exported as part of `B4_results_output.csv`. The Figure (4) is generated from the results obtained. At $\alpha = 0.1$ the storage swings through almost its entire capacity (around 490.68 GWh), reflecting deep seasonal cycling driven by wind and solar variability over the full year. At $\alpha = 0$ the swing reaches 2,484 GWh, confirming that multi-month storage would be required to fully decarbonize the system.

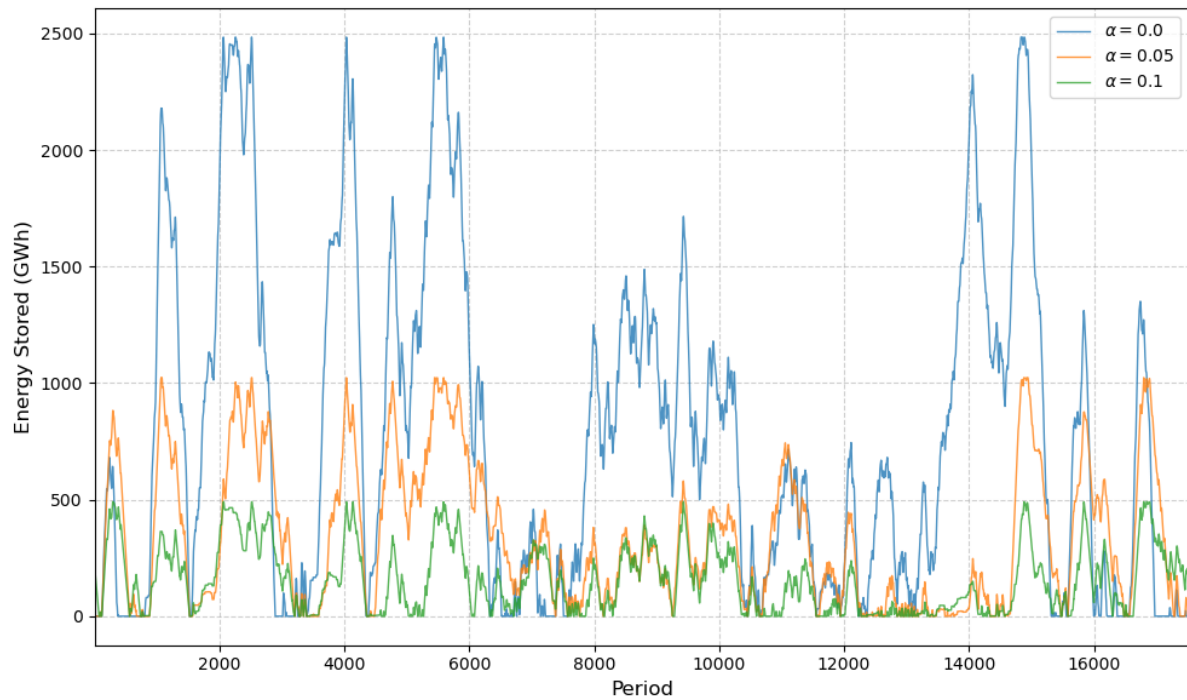
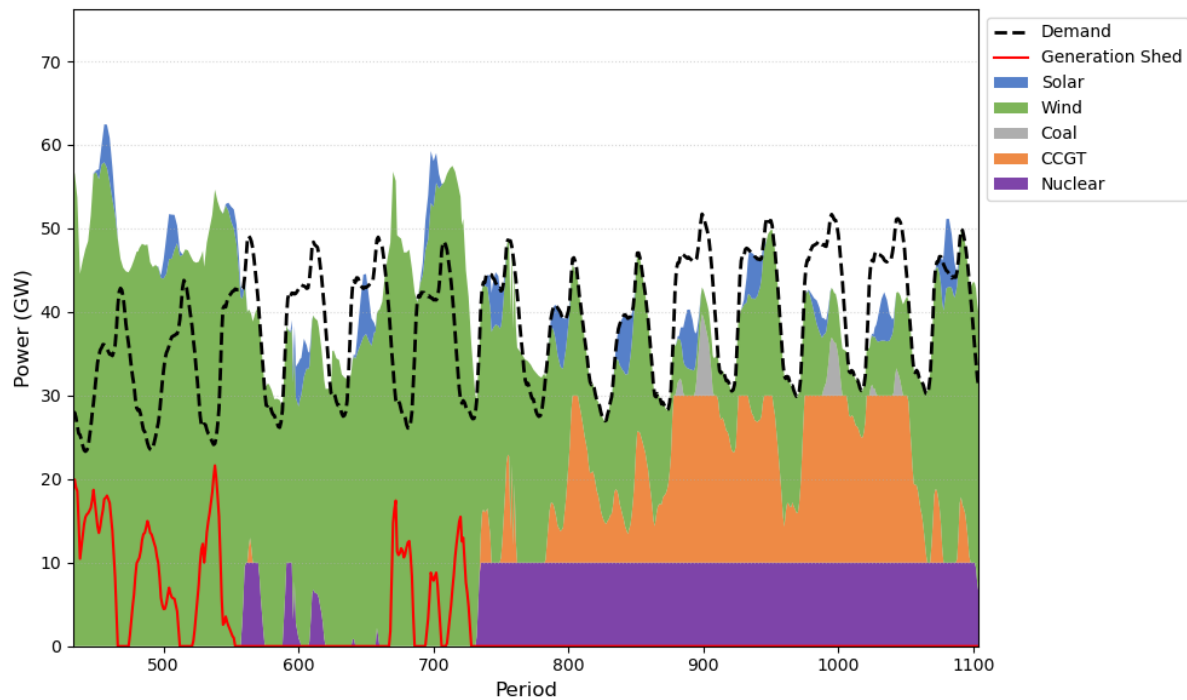


Figure 4: Energy Stored in Electricity Store for the Full Year

Task B5: Slice O dispatch for $\alpha = 0.1$

We extract periods 433 to 1104 (Slice-O) from the $\alpha = 0.1$ full-year solution and produce graphs G1 (stacked generation) as shown in Figure (5), and G2 (storage energy) as shown in Figure (6).

Figure 5: Stacked Generation Power Mix for $\alpha = 0.1$

Compared to Part A, wind now dominates the stack at 58.5 GW installed capacity, frequently exceeding demand and causing significant generation shedding (red line, up to ~ 20 GW) in the early periods 433 to 720. A striking structural shift occurs around **period 730**: Nuclear suddenly appears as a stable 10 GW baseload and CCGT fills the residual, while shedding disappears entirely suggesting the CO_2 constraint drives a fundamentally different dispatch regime in the second half of the slice. Coal is virtually absent throughout, confirming the $\alpha = 0.1$ limit has effectively decarbonized the fossil fleet.

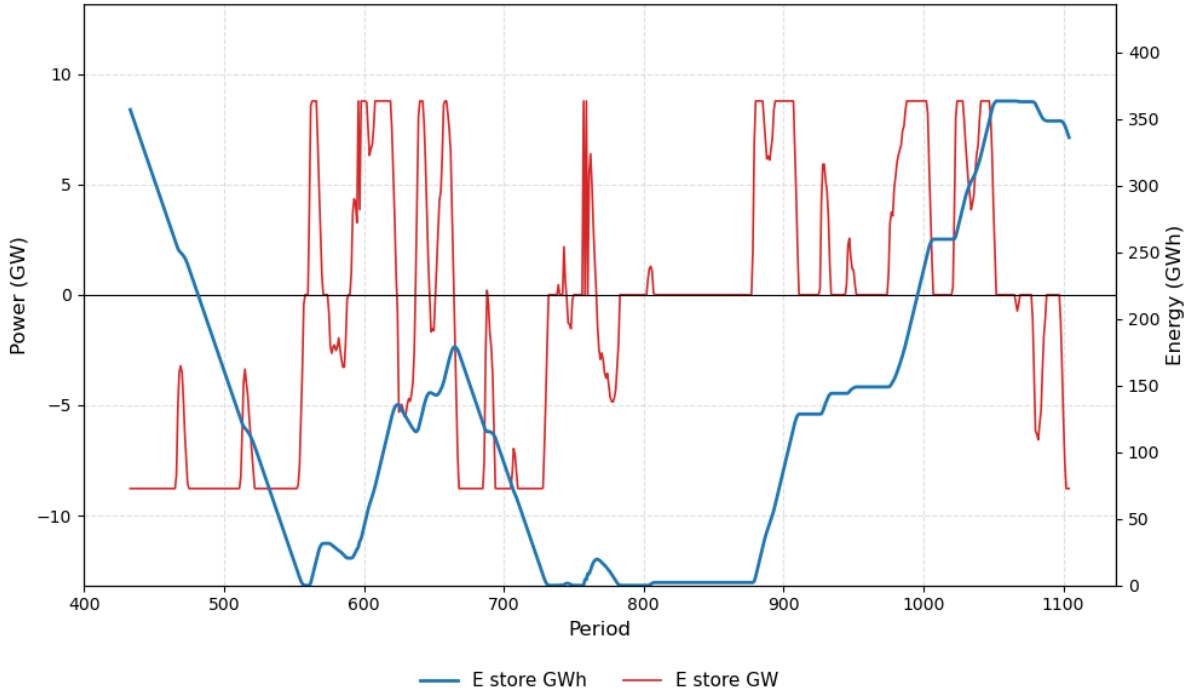


Figure 6: Electricity Store: Power Flow and Energy Stored

The storage profile is dramatically different from Part A, the energy level (blue) swings over a range of ~ 370 GWh, which is large than in Part A, reflecting the much larger optimized PHES. The storage depletes steadily from period 433 to a minimum near period 760, then recharges gradually to ~ 370 GWh by period 1100, indicating seasonal scaling rather than the daily cycling seen in Part A. The net power (red) switches rapidly between charge and discharge at up to ± 8.8 GW, hitting the 8.78 GW capacity bound frequently, showing that the large PHES investment is actively and fully utilized.

Task B6: Comment on Results

The key results from Tasks B1–B5 have been discussed in detail in their respective sections above. The following briefly highlights the most interesting and surprising features.

The most striking result is that $U_{\min} = 0$ is **achieved only by building 758 GW of wind**, which is $37\times$ the lower bound with 1,868,641 GWh of annual curtailment (shedding), revealing that zero emissions is technically feasible but physically unrealistic without constraints on maximum investment or minimum dispatch ability.

Another surprising feature is the **non-monotonic OPEX** across the sweep. OPEX rises at $\alpha = 0.4$ before declining at lower α values, as the CO_2 constraint initially forces expensive CCGT dispatch before sufficient renewable investment makes zero marginal-cost generation dominant.

The **carbon marginal price κ jumps by three orders of magnitude** between $\alpha = 0.05$ (-0.883 £k/tCO₂) and $\alpha = 0$ (-926.1 £k/tCO₂), confirming that the final increment to zero emissions is disproportionately costly compared to all preceding abatement steps.

Additional features to improve realism of the system:

- **Minimum stable generation** for thermal plant, preventing Coal and CCGT from cycling to zero instantaneously.
- **Maximum investment bounds** on wind and solar to reflect land use, planning and grid connection limits.
- **Demand response** allowing load shifting as an alternative to storage investment.
- **Start-up costs and ramp rates** to capture the true operational flexibility constraints of fossil generators.

Task B7: Operation Without Generation Shedding

In the current model, $s_t \geq 0$ (amount of power shed) acts as a slack variable in the power balance constraint, absorbing surplus generation whenever renewable output exceeds demand plus storage capacity. Removing shedding means the power balance becomes a strict equality with no slack, so the model must find alternative ways to manage surplus.

The most direct approach is to **constrain renewable investment** so that generation never exceeds demand plus the storage absorption capacity in any period. However, this would force a tighter CO₂ limit to be achieved through more expensive means such as additional PHES or demand response rather than renewable overbuild.

Alternatively, **increasing PHES energy and power capacity** would allow more surplus to be stored rather than shed. From Table 3, PHES energy capacity already reaches 2,484 GWh at $\alpha = 0$, without shedding it would need to grow further still, making the problem significantly more expensive.

A third option is to **introduce export variables**, allowing surplus generation to be exported rather than curtailed. This effectively replaces the shedding variable with a non-negative export term in the power balance, which is economically equivalent but more physically meaningful.

Without any of these remedies, **removing s_t would render the problem infeasible** at low α values, since no feasible dispatch exists that simultaneously satisfies the power balance, the CO₂ limit, and the fixed demand with the given capacity mix.